

TUAN NGUYEN

Urbana, IL | tuanmn2@illinois.edu | mtuann.github.io | [Google Scholar](https://scholar.google.com/citations?user=mtuann) | github.com/mtuann | [linkedin.com/in/mtuann](https://www.linkedin.com/in/mtuann)
Research interests: efficient machine learning systems, LLM and RAG serving, GPU performance, federated learning

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

M.S. in Computer Science (AI track)

Aug 2025 – May 2027 (expected)

- GPA 3.85/4.00; research on efficient ML systems with [Prof. Fan Lai](#).

VinUniversity (joint program with UIUC)

Hanoi, Vietnam

Ph.D. student in Computer Science

Jan 2023 – Present

- Advisors: [Prof. Kok-Seng Wong](#), [Prof. Khoa Doan](#) (VinUniversity) and [Prof. Fan Lai](#) (UIUC).

VNU University of Engineering and Technology

Vietnam National University, Hanoi

B.S. in Computer Science

2015 – 2019

- High Distinction; ranked 1st of 30.

RESEARCH EXPERIENCE

University of Illinois Urbana-Champaign

Urbana, IL

Research Assistant

Aug 2025 – Present

- Efficient RAG serving: [REVA](#) turns historical LLM attention traces into reusable evidence views, making context compression cheaper at serving time (IEEE ICDM 2026).

VinUniversity — SAIL & VinUni-Illinois Smart Health Center

Hanoi, Vietnam

Research Assistant

Jan 2022 – Jul 2025

- Trustworthy federated learning for healthcare: backdoor attacks and defenses ([NeurIPS 2023](#), [EAAI 2024](#), two first-author preprints), federated unlearning ([ACML 2023](#)) and communication-efficient federated recommendation ([WWW 2024](#)).
- Video-based AI for cardiac function assessment and myocardial-infarction detection in echocardiography ([Frontiers in Cardiovascular Medicine 2023](#)).

VNU University of Engineering and Technology — AI Lab

Hanoi, Vietnam

Research Engineer

Jan 2019 – Dec 2021

- Led a VinIF-funded cardiac-analysis system: left-ventricle detection, ejection-fraction calculation and strain measurement from echocardiography, with an active-learning segmentation pipeline ([KSE 2021](#)).
- Built the Android tablet annotation module and FastAPI review backend that cardiologists used to label 5,000 echocardiography cases; co-inventor on a [patent application](#) published in Vietnam (2022).

PUBLICATIONS

Full list on [Google Scholar](#); * equal contribution.

Conference and journal papers

- Tuan Nguyen**, Qiran Hu, Banruo Liu, Khoa D. Doan, Kok-Seng Wong, Fan Lai. “[REVA: Reusable Evidence View Aggregation for Context-Efficient RAG Serving](#).” *IEEE International Conference on Data Mining (ICDM)*, 2026. [\[code\]](#)
- Ngoc-Hieu Nguyen, Tuan-Anh Nguyen, **Tuan Nguyen**, Vu Tien Hoang, Dung D. Le, Kok-Seng Wong. “[Towards Efficient Communication and Secure Federated Recommendation System via Low-rank Training](#).” *The ACM Web Conference (WWW)*, 2024. [\[code\]](#)
- Thuy Dung Nguyen*, **Tuan Nguyen***, Phi Le Nguyen, Hieu H. Pham, Khoa D. Doan, Kok-Seng Wong. “[Backdoor Attacks and Defenses in Federated Learning: Survey, Challenges and Future Research Directions](#).” *Engineering Applications of Artificial Intelligence (EAAI)*, 2024. [\[code\]](#)
- Thuy Dung Nguyen, **Tuan Nguyen**, Anh Tran, Khoa D. Doan, Kok-Seng Wong. “[IBA: Towards Irreversible Backdoor Attacks in Federated Learning](#).” *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. [\[code\]](#)
- Thai-Hung Nguyen, Hong-Phuc Vu, Dung Thuy Nguyen, **Tuan Nguyen**, Khoa D. Doan, Kok-Seng Wong. “[An Empirical Study of Federated Unlearning: Efficiency and Effectiveness](#).” *Asian Conference on Machine Learning (ACML)*, 2023. [\[code\]](#)

- [6] **Tuan Nguyen***, Phi Nguyen*, Dai Tran, Hung Pham, et al. “[Ensemble Learning of Myocardial Displacements for Myocardial Infarction Detection in Echocardiography.](#)” *Frontiers in Cardiovascular Medicine*, 2023. [code]
- [7] Thuy Hang Duong Thi, **Tuan Nguyen**, Phi Nguyen Van, Quoc Long Tran. “[Fully Automated Machine Learning Pipeline for Echocardiogram Segmentation.](#)” *International Conference on Knowledge and Systems Engineering (KSE)*, 2021.

Preprints

- [8] **Tuan Nguyen**, Sze Jue Yang, Khoa D. Doan, Chee Seng Chan, Kok-Seng Wong. “[FLAT: Revealing Hidden Latent-Conditioned Backdoor Failures in Federated Learning.](#)” *arXiv:2508.04064*, 2025. [code]
- [9] **Tuan Nguyen**, Dung Thuy Nguyen, Khoa D. Doan, Kok-Seng Wong. “[Non-Cooperative Backdoor Attacks in Federated Learning: A New Threat Landscape.](#)” *arXiv:2407.07917*, 2024.

Patent

- [10] Tran Quoc Long, Le Sy Vinh, Le Tuan Thanh, Nguyen Van Phi, Le Pham Van Linh, **Tuan Nguyen**, Tran Minh Duc. “[Automated Quantification of Ejection Fraction Directly from Echocardiography Video.](#)” *Patent application, National Office of Intellectual Property of Vietnam*, 2022.

SELECTED PROJECTS

CUDA convolution kernels (UIUC CS 483 course project): fused implicit-GEMM kernels for CNN layers 2026
Federated-learning framework: modular PyTorch framework for FL attack and defense experiments | [code](#)
Research Papers Explorer: search and filter curated paper collections | [app](#), [code](#)

TEACHING

Teaching Assistant, VinUniversity: Intro to Programming, OOP & Data Structures, Cybersecurity 2022 – 2025
Teaching Assistant, VNU-UET: Intro to Programming Languages, Advanced Programming 2019 – 2021

SERVICE

Reviewer: NeurIPS (2024–2026), ICLR (2024–2026), ICML (2025–2026), AISTATS (2026)
 NeurIPS 2025 Datasets & Benchmarks track; workshops on backdoors (NeurIPS 2023) and DIG-BUG (ICML 2025)

HONORS & AWARDS

Vingroup Scholarship for M.S. study at UIUC	2025
Finalist & Demo Showcase (top 8 of 900 teams), K2 Think Hackathon	2025
Full scholarship for Ph.D. study, VinUniversity	2023
Second Prize, National Mathematics Contest for University Students	2019
Young Face Award, Vietnam National University	2019
VNU Mitsubishi Scholarship (MUFG Bank)	2018
Second Prize, ACM ICPC Vietnam National Programming Contest	2017
Second Prize, National Informatics Contest for High School Students	2015